

BANK CUSTOMER CHURN ANALYSIS REPORT

Exploratory Data Analysis (EDA)

1. PROJECT TITLE

Bank Customer Churn Prediction using Exploratory Data Analysis

2. OBJECTIVE

The objective of this project is to analyze bank customer data and identify key factors that influence customer churn. The analysis helps in understanding customer behavior and provides insights for improving customer retention strategies.

3. DATASET DESCRIPTION

The dataset contains information about bank customers including demographic details, account information, and financial behavior.

Features in dataset:

- customer_id → Unique customer identifier
- credit_score → Credit score of customer
- country → Customer's country
- gender → Gender of customer
- age → Age of customer
- tenure → Years with bank
- balance → Account balance
- products_number → Number of bank products used
- credit_card → Whether customer has credit card (1/0)
- active_member → Whether customer is active (1/0)
- estimated_salary → Estimated income

- churn → Target variable (1 = exited, 0 = retained)
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4. DATA CLEANING

The dataset was preprocessed to ensure quality analysis:

- Removed duplicate records
 - Checked and confirmed no missing values
 - Verified data types of all columns
 - Ensured consistency in categorical values
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5. EXPLORATORY DATA ANALYSIS

5.1 Customer Churn Distribution

The dataset shows imbalance between retained and churned customers:

- Majority customers are retained (0)
 - Around 20% customers have churned (1)
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5.2 Age Analysis

- Middle-aged and older customers show higher churn probability
 - Younger customers are more likely to stay active
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5.3 Balance Analysis

- Customers with low or zero balance are more likely to churn
 - High balance customers tend to stay longer
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5.4 Credit Score Analysis

- Customers with lower credit scores have higher churn rates
- Credit score plays an important role in customer retention

5.5 Geography Analysis

- Customers from certain countries show higher churn rates
- Regional differences significantly affect customer behavior

5.6 Gender Analysis

- Gender has minimal impact on churn behavior
- Both male and female customers show similar churn patterns

5.7 Active Membership Analysis

- Inactive members are more likely to churn
- Active engagement reduces churn probability

6. CORRELATION ANALYSIS

Correlation analysis shows:

- Age is positively correlated with churn
- Balance has a moderate relationship with churn
- Active membership is negatively correlated with churn
- Credit score has weak to moderate influence

7. KEY INSIGHTS

- Customers with low balance are at higher risk of churn
- Age is an important factor in predicting churn
- Active members are less likely to leave the bank
- Credit score impacts customer retention behavior
- Geography plays a significant role in churn distribution

8. 💡 BUSINESS RECOMMENDATIONS

- Focus on retaining low-balance customers with offers
- Improve engagement programs for inactive users
- Introduce loyalty rewards for long-tenure customers
- Provide targeted services based on geography
- Monitor high-risk customers using predictive analytics

9. 📌 CONCLUSION

The analysis clearly shows that customer churn is influenced by financial behavior, activity level, age, and geography. These insights can help banks design better customer retention strategies and reduce customer loss effectively.

10. 🛠️ TOOLS USED

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Jupyter Notebook